

Limit laws (Only for finite limits)

$$1. \lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$2. \lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$3. \lim_{x \rightarrow a} (c f(x)) = c \lim_{x \rightarrow a} f(x)$$

$$4. \lim_{x \rightarrow a} f(x) \cdot g(x) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$5. \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

$$6. \lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x) \right)^n$$

$$7. \lim_{x \rightarrow a} c = c$$

$$8. \lim_{x \rightarrow a} x = a$$

$$9. \lim_{x \rightarrow a} x^n = a^n$$

$$10. \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$$

$$11. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

Examples:

1. $\lim_{x \rightarrow 2} (3x^2 + 4 \sin \frac{\pi}{x} + 5)$

$$= \lim_{x \rightarrow 2} 3x^2 + \lim_{x \rightarrow 2} 4 \sin \frac{\pi}{x} + \lim_{x \rightarrow 2} 5$$
$$= 3 \lim_{x \rightarrow 2} x^2 + 4 \lim_{x \rightarrow 2} \sin \frac{\pi}{x} + 5$$
$$= 3 \times 4 + 4 \times 1 + 5 = 21$$

2. $\lim_{x \rightarrow 4} \sqrt{4-x^2} = \sqrt{\lim_{x \rightarrow 4} (4-x^2)}$ does not exist

$$\lim_{x \rightarrow 1} \sqrt{4-x^2} = \sqrt{3}$$

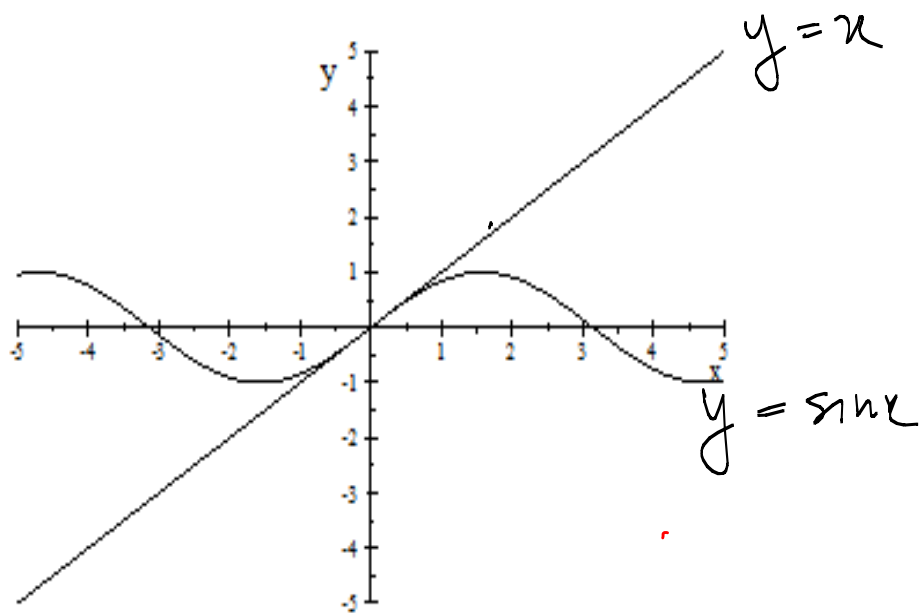
Limits of the form $\left(\frac{0}{0}\right)$ ← Make a plan.

1. Factorising : $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = 4$ $\left(\frac{0}{0}\right)$

2. Multiplying by the conjugate

$$\lim_{x \rightarrow 0} \frac{x}{\sqrt{x+4} - 2} \times \frac{\sqrt{x+4} + 2}{\sqrt{x+4} + 2}$$
$$= \lim_{x \rightarrow 0} \frac{x(\sqrt{x+4} + 2)}{\cancel{x+4} - 4}$$
$$= \lim_{x \rightarrow 0} (\sqrt{x+4} + 2) = 4$$

3. Using $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$ $\leftarrow$ NB.



$$a. \lim_{x \rightarrow 0} \frac{\sin(4x)}{4x} = 4 \lim_{4x \rightarrow 0} \frac{\sin 4x}{4x} = 4. \quad \left| \begin{array}{l} x \rightarrow 0 \\ \Rightarrow \\ 4x \rightarrow 0 \end{array} \right.$$

$$\lim_{x \rightarrow 0} \frac{\sin x}{3x} = \frac{1}{3} \lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{1}{3}.$$

$$b. \lim_{x \rightarrow 0} \frac{\tan x}{x} = \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{1}{\cos x}$$

$\lim_{x \rightarrow 0} \frac{\cos x}{x} \neq 0$
Later.

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \lim_{x \rightarrow 0} \frac{1}{\cos x} = 1,$$

$$c. \quad \lim_{x \rightarrow 0} \frac{\sin 5x}{\sin 3x}$$

$$= \frac{5}{3} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x} \times \frac{3x}{\sin 3x}$$

$$= \frac{5}{3} \underbrace{\lim_{5x \rightarrow 0} \frac{\sin 5x}{5x}}_1 \times \underbrace{\lim_{3x \rightarrow 0} \frac{3x}{\sin 3x}}_1 = \frac{5}{3}$$

$$d. \quad \lim_{x \rightarrow 0} \frac{\tan x^2}{x}$$

$$= \lim_{x \rightarrow 0} \frac{\sin x^2}{x^2} \times \frac{x}{\cos x^2}$$

$$= \underbrace{\lim_{x^2 \rightarrow 0} \frac{\sin x^2}{x^2}}_1 \times \underbrace{\lim_{x \rightarrow 0} \frac{x}{\cos x^2}}_1 = 1$$

(Note: In the original image, there are arrows indicating $x \rightarrow 0$ for the x in the numerator and $\cos x^2 \rightarrow 1$ for the denominator in the second limit.)

$$= 1$$

$$\lim_{x \rightarrow 0} \frac{x}{\sin x}$$

$$= \lim_{x \rightarrow 0} \frac{1}{\frac{\sin x}{x}}$$

$$= \frac{1}{\lim_{x \rightarrow 0} \frac{\sin x}{x}}$$

$$= 1$$

Limits involving zero

Forms $\left(\frac{c}{d}\right)$ $\left(\frac{0}{0}\right)$ $\left(\frac{c}{0}\right)$ $\left(\frac{0}{c}\right)$

$$\left(\frac{c}{d}\right) : 1. \lim_{x \rightarrow 2} \frac{x^2 - 3}{x + 2} = \frac{1}{4} \quad \left(\frac{1}{4}\right)$$

$$2. \lim_{x \rightarrow \pi/2} \frac{\sin x}{x} = \frac{2}{\pi} \quad \left(\frac{1}{\pi/2}\right)$$

$\left(\frac{0}{0}\right) \sim$ factorisation, conjugate, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$,
half limits eg $\frac{|x|}{x}$

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x} \times \frac{1 + \cos x}{1 + \cos x} \quad \left(\frac{0}{0}\right)$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x}{x(1 + \cos x)} \rightarrow \sin^2 x$$

$$= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{\sin x}{1 + \cos x} = \underbrace{\lim_{x \rightarrow 0} \frac{\sin x}{x}}_1 \times \underbrace{\lim_{x \rightarrow 0} \frac{\sin x}{1 + \cos x}}_0$$

$\left(\frac{c}{0}\right)$: vertical asymptotic behaviour
Check half limits (from left and from right)

$$1. \lim_{x \rightarrow 4} \frac{x^2}{x - 4} \quad \left(\frac{c}{0}\right)$$

$$\lim_{x \rightarrow 4^+} \frac{x^2}{x-4} = \infty \quad \left(\frac{16}{0^+} \right)$$

$$\lim_{x \rightarrow 4^-} \frac{x^2}{x-4} = -\infty \quad \left(\frac{16}{0^-} \right)$$

$\Rightarrow x=4$ is a VA.

3. $\lim_{x \rightarrow 0} \frac{\cos x}{x} \quad \left(\frac{c}{0} \right)$

$$\lim_{x \rightarrow 0^+} \frac{\cos x}{x} = \infty$$

$$\lim_{x \rightarrow 0^-} \frac{\cos x}{x} = -\infty$$

$\Rightarrow x=0$ is a VA.

$\left(\frac{0}{c} \right)$: Limit value is zero

1. $\lim_{x \rightarrow 2} \frac{x-2}{x} = 0 \quad \left(\frac{0}{2} \right)$

2. $\lim_{x \rightarrow 0} \frac{x}{\cos x} = 0 \quad \left(\frac{0}{1} \right)$

Squeeze theorem

If $f(x) \leq g(x) \leq h(x)$ near a
(except perhaps for $x=a$) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then $\lim_{x \rightarrow a} g(x) = L$.

$$\lim_{x \rightarrow 0} x^2 \sin \frac{1}{x}$$

$$-1 \leq \sin \frac{1}{x} \leq 1$$

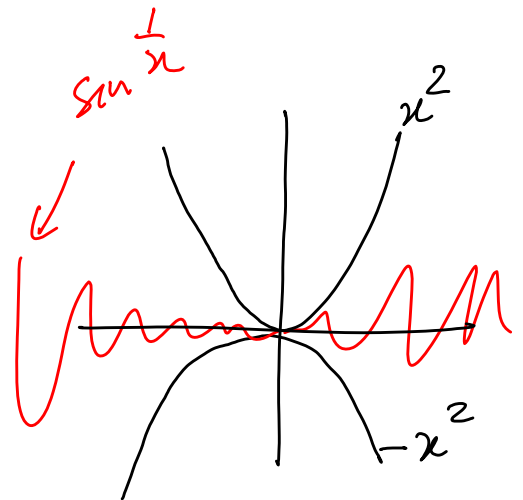
$$\Rightarrow -x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$$

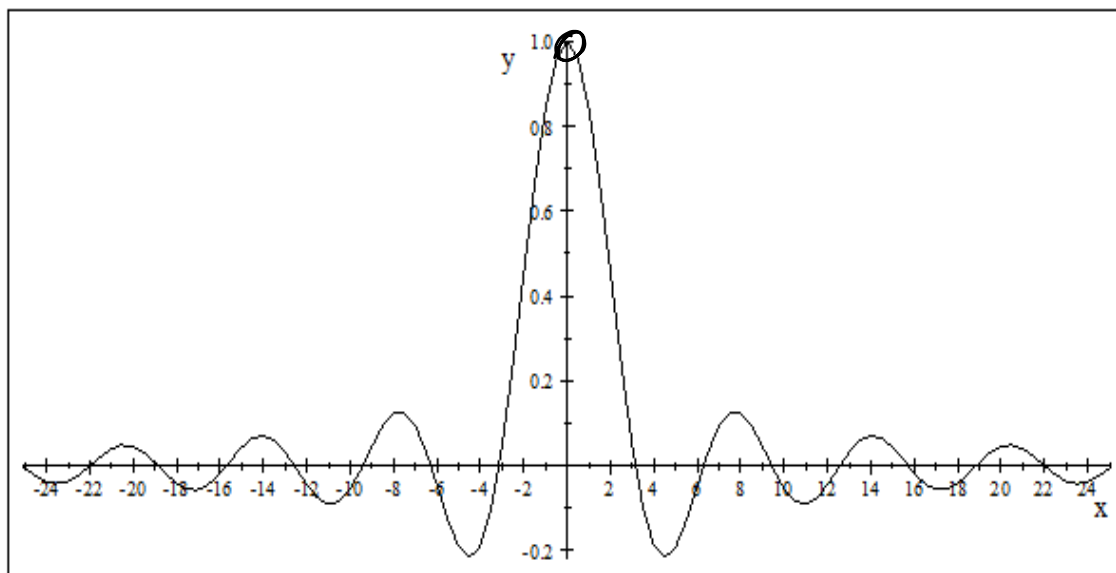
$$\lim_{x \rightarrow 0} (-x^2) = 0$$

$$\lim_{x \rightarrow 0} (x^2) = 0$$

Because $-x^2 \leq x^2 \sin \frac{1}{x} \leq x^2$ we can say

$$\text{that } \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0.$$





$$e^{a+ib} = e^a (\cos b + i \sin b) \quad \left\{ \begin{array}{l} \text{Definition.} \\ \end{array} \right.$$

$$e^{i\pi} = e^{0+i\pi} = e^0 (\cos \pi + i \sin \pi) \\ = -1$$

$$\Rightarrow e^{i\pi} + 1 = 0.$$